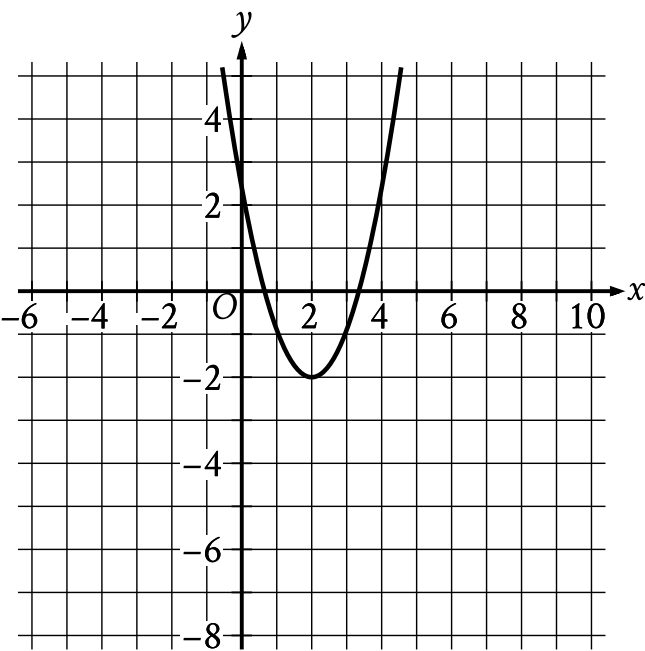


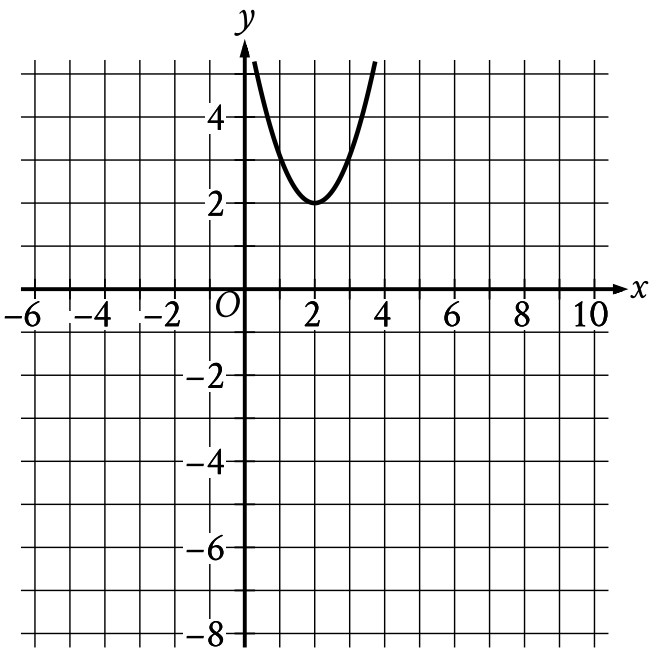
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

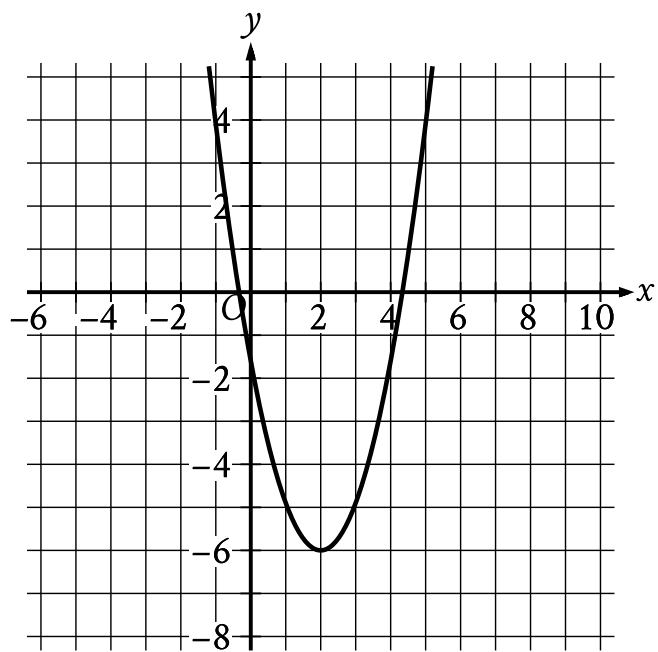


The graph shown will be translated up 4 units. Which of the following will be the resulting graph?

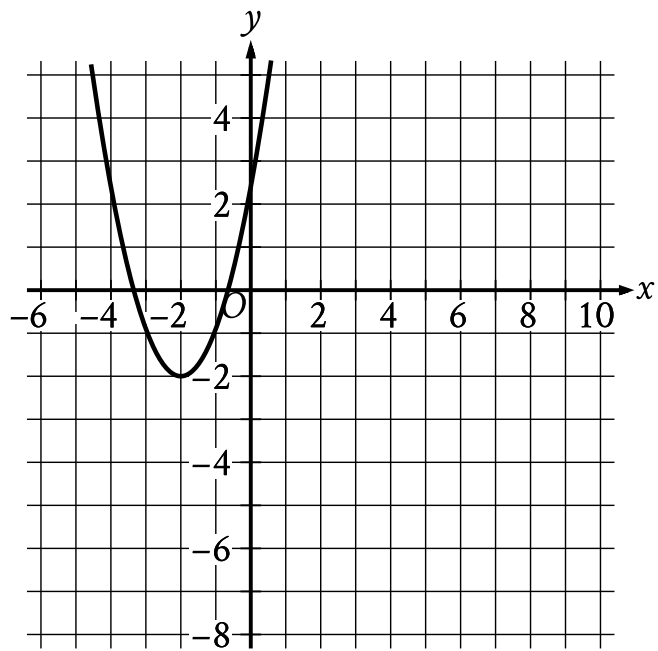
Answer



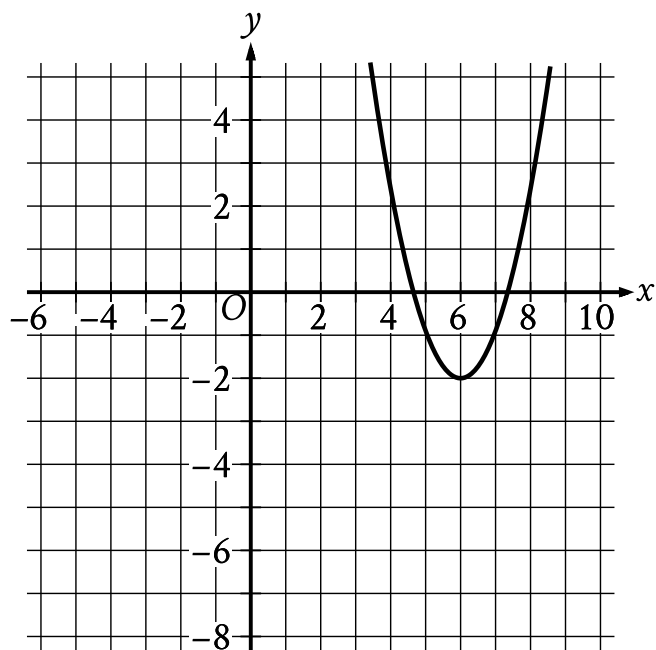
A.



B.



C.



D.

Correct Answer: A

Rationale

Choice A is correct. When a graph is translated up **4** units, each point on the resulting graph is **4** units above the point on the original graph. In other words, the y -value of each point on the graph increases by **4**. The graph shown passes through the points $(1, -1)$, $(2, -2)$, and $(3, -1)$. It follows that when the graph shown is translated up **4** units, the resulting graph will pass through the points $(1, -1 + 4)$, $(2, -2 + 4)$, and $(3, -1 + 4)$. These points are $(1, 3)$, $(2, 2)$, and $(3, 3)$, respectively. Of the given choices, only the graph in choice A passes through the points $(1, 3)$, $(2, 2)$, and $(3, 3)$.

Choice B is incorrect. This is the result of translating the graph down, rather than up, **4** units.

Choice C is incorrect. This is the result of translating the graph left, rather than up, **4** units.

Choice D is incorrect. This is the result of translating the graph right, rather than up, **4** units.